Responsive Design and Accessibility Implementation

Project Name: Studli – Study Planner App
Week: 4
Prepared by: Smriti
Project tool: Figma

1. Project Overview

Studli is a student-focused study planner application designed to help students organise their academic activities and manage their study routines effectively. The application provides features such as a home dashboard, task planning, progress tracking, profile management and study partner connectivity.

The purpose of this project is to improve the existing Studli interface by implementing responsive design principles and accessibility improvements across mobile, tablet and desktop devices.

The original interface was primarily designed for mobile screens. Therefore, the project focuses on adapting the existing layouts to different screen dimensions while maintaining consistency in the application's visual identity, including its purple and lavender colour palette, typography, icons, cards and overall content structure.

Using Figma, responsive mockups were developed for three device categories:

- **Mobile: 430 × 932 px**
- **Tablet: 1024 × 768 px**
- **Desktop: 1440 × 900 px**

The project also focuses on accessibility improvements, including text contrast, readable typography, appropriately sized interactive elements and visible keyboard focus indicators.

The overall objective is to create a consistent, accessible and user-friendly interface that provides a suitable experience across different screen sizes without changing the original purpose or functionality of the application.

2. Project Objectives

The primary objective of the Studli project was to improve the existing application interface by implementing responsive design and accessibility considerations.

The specific objectives were:

1. Responsive Adaptability: Adapt the existing mobile interface to tablet and desktop screen sizes while ensuring that the layouts remain organised and usable.

2. Improved User Experience: Arrange interface elements, navigation and content in a way that makes the application easier to use across different devices.

3. Consistent Visual Identity: Preserve the original Studli colour palette, typography, icons and overall design style throughout the redesign.

4. Accessible Interface Design: Improve text readability, colour contrast, touch target dimensions and visibility of keyboard focus indicators.

5. Effective Space Utilisation: Make appropriate use of available screen space through suitable content arrangements, spacing and navigation layouts.

6. Typography Adaptation: Adjust font sizes and text hierarchy to maintain readability across different screen dimensions.

7. Design Documentation: Prepare responsive mockups and document the design decisions, specifications and accessibility improvements.

The overall goal was to create a consistent, responsive and accessible Studli interface without changing its original visual identity.

3. Review of Original Design

The existing Studli interface was reviewed before beginning the responsive redesign. The original design consisted of seven mobile screens representing the main features of the application:

1. Home Dashboard
2. My Planner
3. My Progress
4. My Profile
5. Study Partner
6. Add a Task
7. Add a Task – Entered

Existing Design Strengths

The original interface had a consistent visual identity, supported by a purple and lavender colour palette, rounded cards, simple icons and a clear content hierarchy. The screens presented information in an organised manner, making the main application features easy to identify.

Areas Identified for Improvement

The original layouts were primarily structured for mobile screen dimensions. Adapting them to larger screens required changes to content arrangement, navigation placement, spacing and card positioning.

The interface also required an accessibility review, particularly for text contrast, interactive element sizes and keyboard focus visibility.

Design Approach

The redesign retained the original content, colours, typography and visual elements wherever possible. Instead of creating a completely new interface, the focus was placed on improving responsiveness and accessibility while preserving the original design identity.

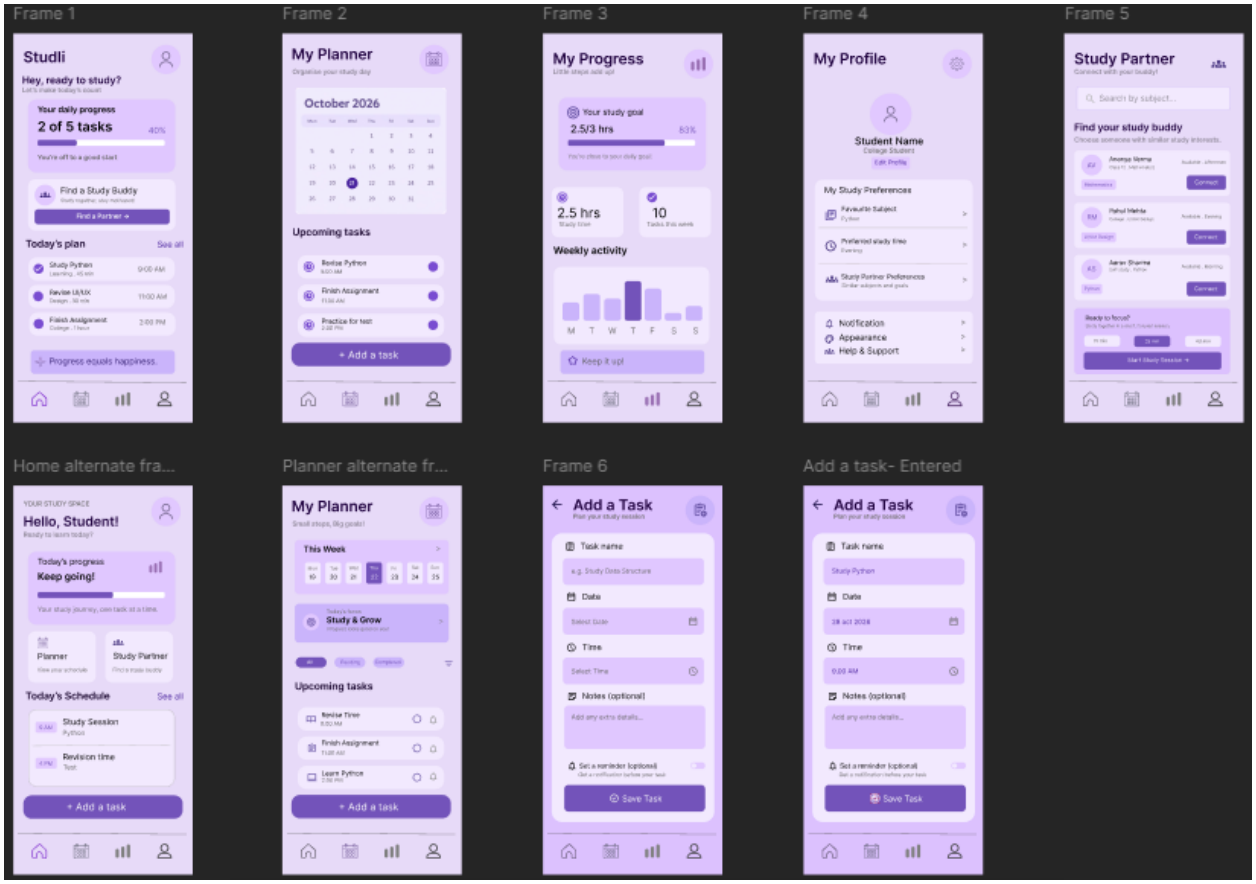


Figure 1: *Original Studli mobile interface before responsive redesign.*

4. Responsive Design Research and Principles

Responsive design plays an important role in creating interfaces that provide a consistent experience across devices with different screen dimensions. It involves adjusting the arrangement, spacing and presentation of interface elements according to the available screen space.

For the Studli redesign, several design principles were considered to guide the adaptation of the existing interface.

1. Screen Adaptability

The interface should adjust to different screen dimensions without making content appear crowded, stretched or difficult to navigate.

2. Layout Organisation

Content should be arranged according to the available space. Single-column layouts are suitable for smaller screens, while wider screens can accommodate multiple content sections.

3. Flexible Spacing and Alignment

Consistent spacing, alignment and element positioning help maintain a balanced interface across different device sizes.

4. Auto Layout and Constraints

Figma's Auto Layout and constraint features help maintain organised component structures and control how elements respond when frame dimensions change.

5. Breakpoint-Based Adaptation

Specific screen-width ranges are used to determine when navigation, card arrangements and content layouts should change.

6. Readability and Visual Hierarchy

Text, headings and important interface elements should remain clearly distinguishable. Proper typography, spacing and contrast support easier navigation and content understanding.

7. Design Consistency

The application's original colour palette, typography, icons and component styles should remain consistent across all device versions.

These principles formed the foundation of the Studli responsive redesign. They helped guide layout decisions and maintain a balance between visual consistency, usability and adaptability.

5. Responsive Design Specification

Responsive design specifications were established to define how the Studli interface would adapt to different device dimensions. Separate frame sizes, layout arrangements and navigation patterns were selected for mobile, tablet and desktop versions.

5.1 Device Dimensions

The following frame dimensions were used during the redesign:

Device	Width	Height	Orientation
Mobile	430 px	932 px	Portrait
Tablet	1024 px	768 px	Landscape
Desktop	1440 px	900 px	Landscape

5.2 Responsive Breakpoints

The following screen-width ranges were used to organise the responsive layouts:

- **Mobile:** Below 768 px
- **Tablet:** 768–1024 px
- **Desktop:** Above 1024 px

These breakpoints provide a reference for adapting the interface structure and navigation according to the available screen width.

5.3 Navigation Specifications

- **Mobile:** A bottom navigation bar was used to provide convenient access to the main application sections.
- **Tablet:** A vertical sidebar was introduced to organise navigation within the landscape-oriented layout.
- **Desktop:** The vertical sidebar was retained to provide consistent navigation and a structured interface.

5.4 Content Layout Specifications

- **Mobile:** A single-column arrangement was used to organise content within the limited screen width.
- **Tablet:** Content sections were organised using wider layouts and two-column arrangements where suitable.
- **Desktop:** Wider content areas and spacious arrangements were used to accommodate multiple sections comfortably.

5.5 Visual Design Specifications

The original purple and lavender colour palette, typography, icons, rounded cards, buttons and existing content were retained across all three device versions.

The specifications establish a consistent visual structure while allowing the interface layout to adapt to different screen dimensions.

6. Mobile Design Implementation

The mobile version of Studli was designed using a frame size of 430 × 932 px in portrait orientation. The layout follows a single-column structure to keep information organised and accessible within the available screen space.

The bottom navigation bar provides convenient access to the main sections of the application. Cards, task information and other content sections are arranged vertically to maintain a clear visual hierarchy.

6.1 Screen Implementations

The following seven screens were adapted for the mobile layout:

1. Home Dashboard: Displays daily progress, planner information and study partner access.
2. My Planner: Organises scheduled tasks and calendar information.
3. My Progress: Presents study statistics and weekly activity.
4. My Profile: Displays profile information and study preferences.
5. Study Partner: Provides access to study partner information and session options.
6. Add a Task: Allows users to enter task details.
7. Add a Task – Entered: Displays the task form with entered information.

6.2 Design Considerations

The mobile redesign focused on maintaining readable content, consistent spacing, clear navigation and appropriately sized interactive elements.

The original colours, typography, icons and content were preserved to maintain consistency with the existing Studli interface.

All seven screens were reviewed at different mobile widths to check layout alignment, content positioning and responsiveness.

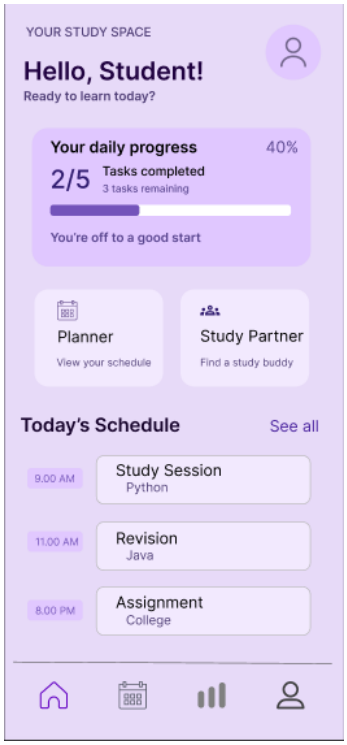


Figure 2: *Mobile Home Frame*

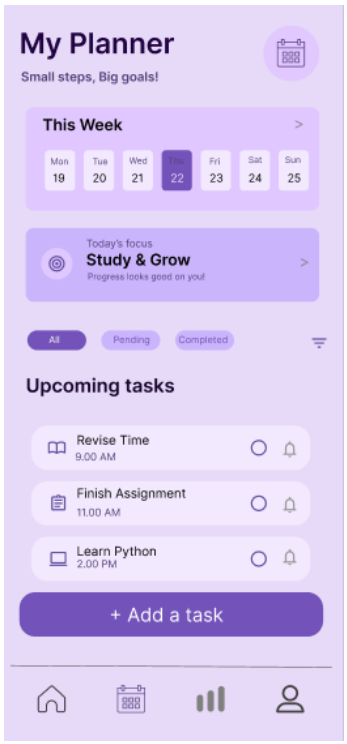


Figure 3: *Mobile Planner Frame*

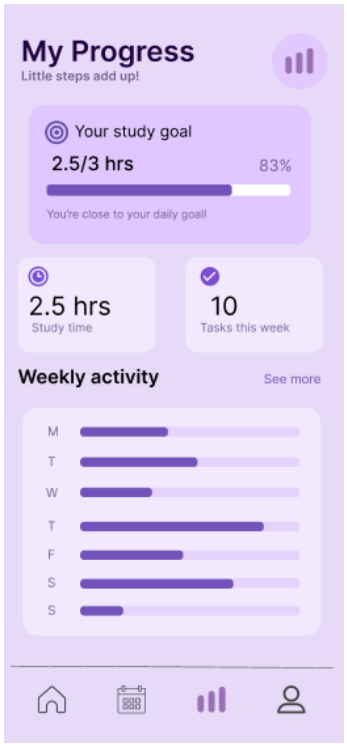


Figure 4: *Mobile Progress Frame*

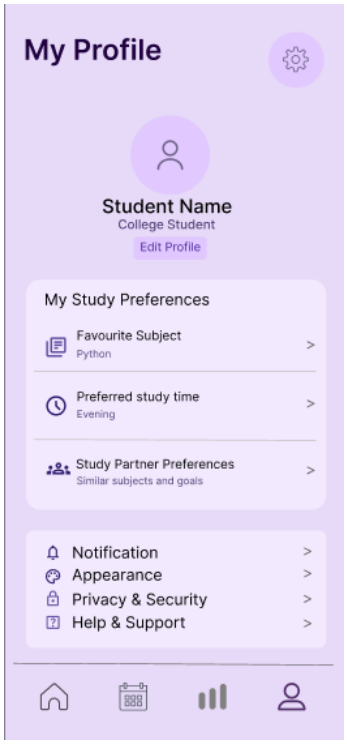


Figure 5*: Mobile Profile Frame*

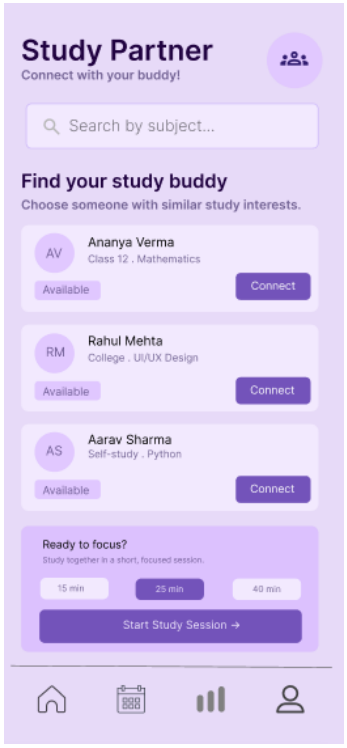


Figure 6: *Mobile Study Partner Frame*

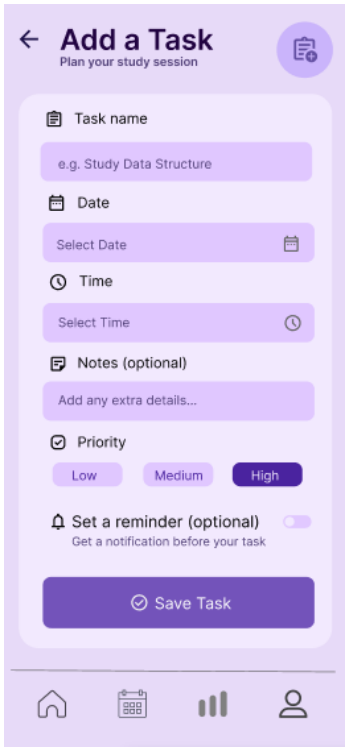


Figure 7: *Mobile Add a Task Frame*

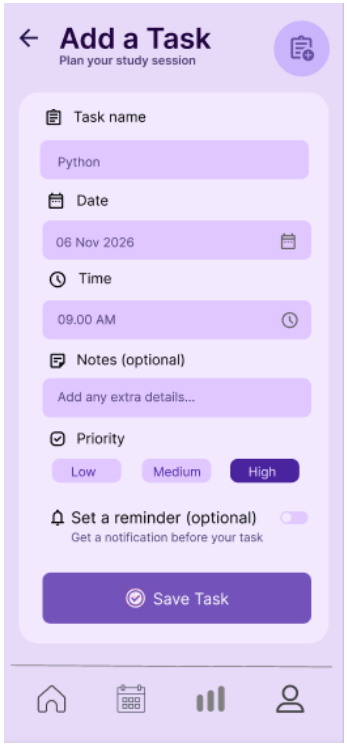


Figure 8: *Mobile Add a task- Entered Frame*

7. Tablet Design Implementation

The tablet version of Studli was designed using a frame size of 1024 × 768 px in landscape orientation. The layout was adapted to make effective use of the wider screen space while maintaining the original visual identity of the application.

The mobile bottom navigation was replaced with a vertical sidebar to provide structured navigation. The main content area was organised into wider sections and two-column arrangements where suitable.

7.1 Screen Implementations

Five main application screens were adapted for the tablet layout:

1. Home Dashboard: Organises daily progress, planner information and study partner access.

2. My Planner: Presents calendar information and scheduled tasks.

3. My Progress: Displays study statistics and activity information.

4. My Profile: Organises personal information, study preferences and settings.

5. Study Partner: Presents study partner information and session options.

7.2 Design Considerations

The tablet redesign focused on using the available horizontal space effectively. A vertical sidebar was introduced, while the main content was arranged using wider layouts and two-column structures where appropriate.

The original colour palette, typography, icons, buttons and existing content were preserved to maintain consistency with the mobile version.

All five tablet screens were reviewed for alignment, spacing, content positioning and responsiveness.

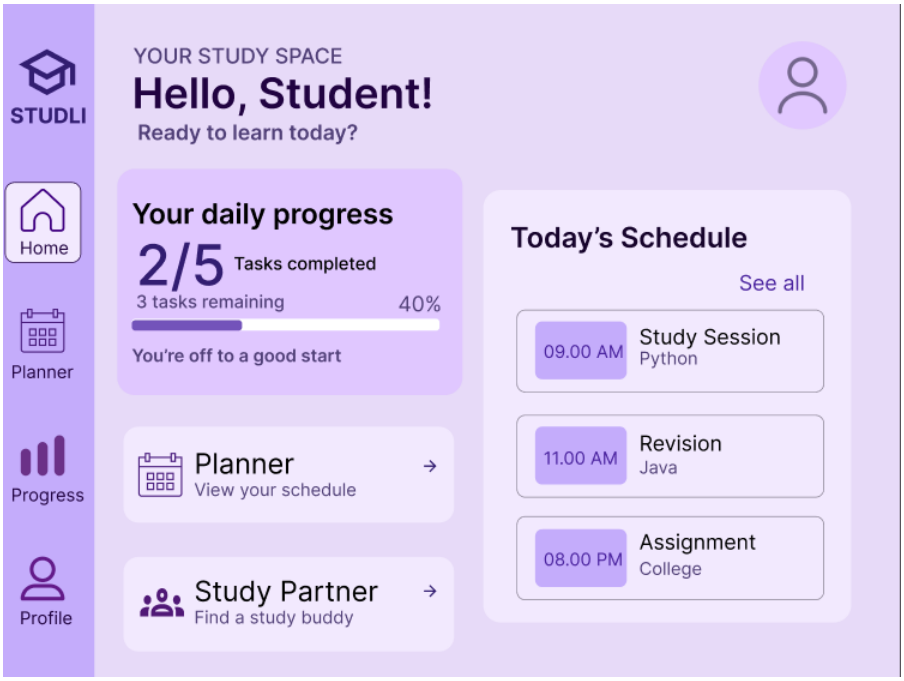


Figure 9: *Tablet Home Frame*

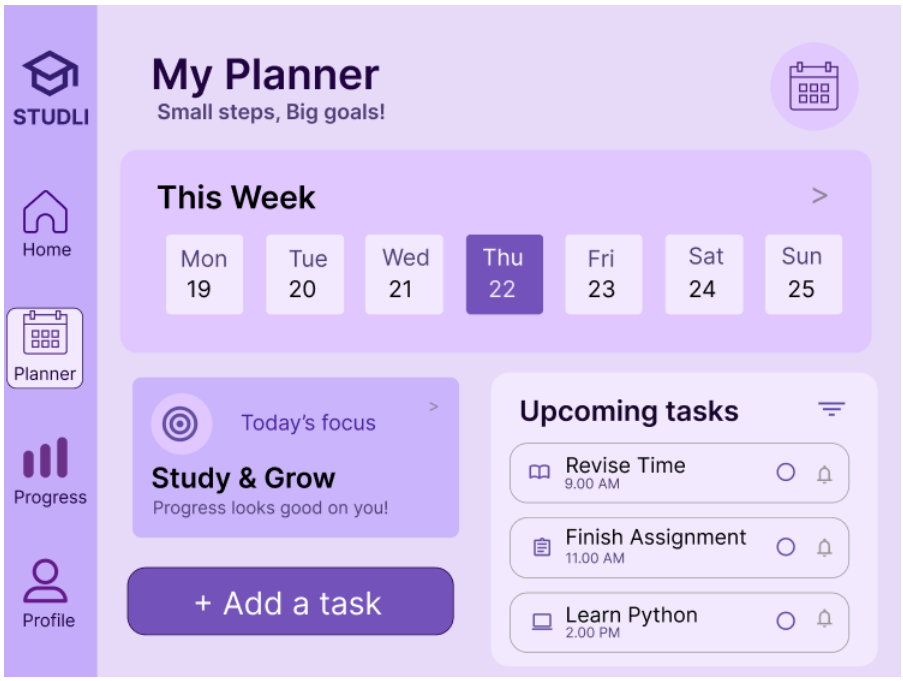


Figure 10: *Tablet Planner Frame*

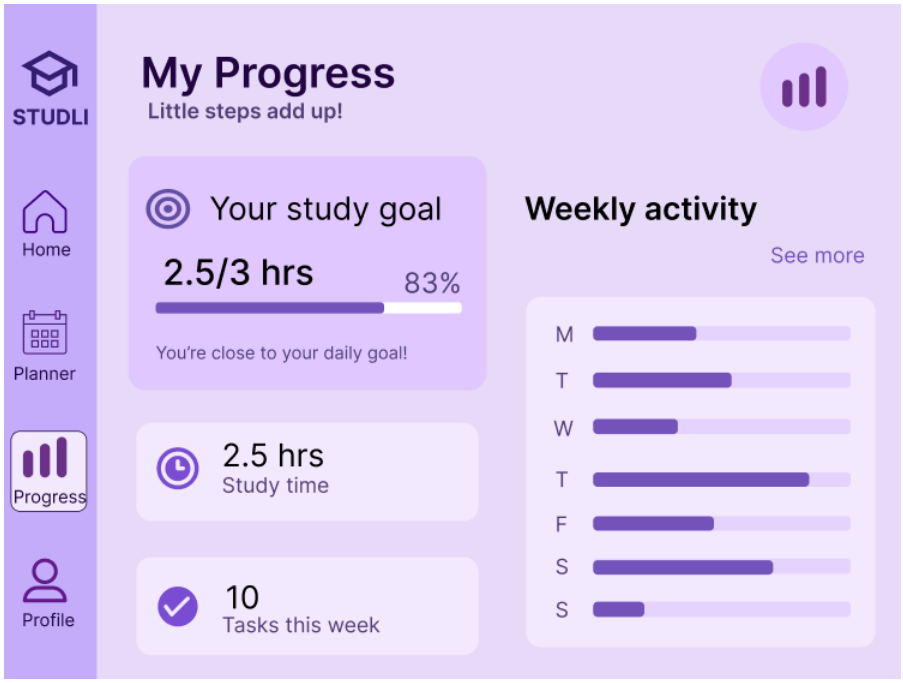


Figure 11: *Tablet Progress Frame*



Figure 12: *Tablet Profile Frame*

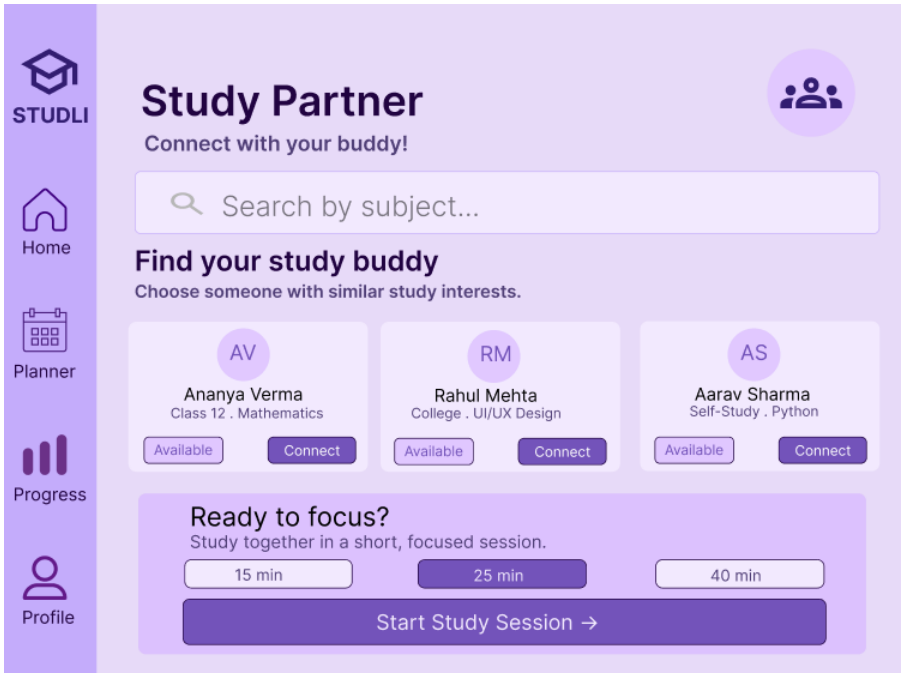


Figure 13: *Tablet Study Partner Frame*

8. Desktop Design Implementation

The desktop version of Studli was designed using a frame size of 1440 × 900 px in landscape orientation. The layout was developed to utilise the larger screen area while maintaining a clear and organised interface.
A vertical sidebar was used for navigation, providing consistent access to the main application sections. The main content area was given more horizontal space to accommodate wider cards, multiple content sections and improved spacing.

8.1 Screen Implementations

Five main application screens were adapted for the desktop layout:
1. Home Dashboard: Displays daily progress, planner information and study partner access.
2. My Planner: Organises calendar information and scheduled tasks.
3. My Progress: Presents study statistics and weekly activity.
4. My Profile: Displays personal information, study preferences and settings.
5. Study Partner: Provides study partner information and session options.

8.2 Design Considerations

- The desktop redesign focused on effective use of available screen space, consistent navigation and clear content organisation.
- Wider content areas and spacious arrangements were used to improve the presentation of information without overcrowding the interface.
- The original purple and lavender colour palette, typography, icons, buttons and existing content were retained to maintain visual consistency across mobile, tablet and desktop versions.
- All five desktop screens were reviewed for alignment, spacing, content positioning and responsiveness.

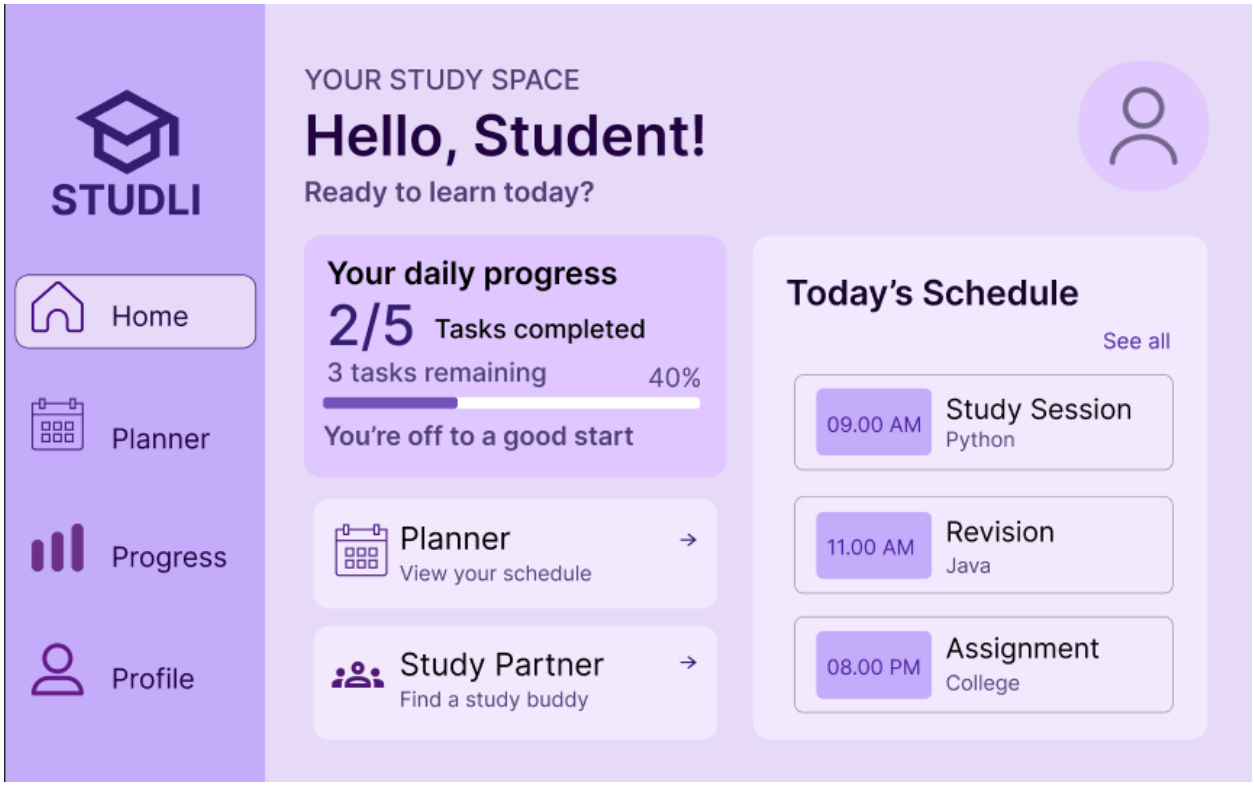


Figure 14: *Desktop Home Frame*

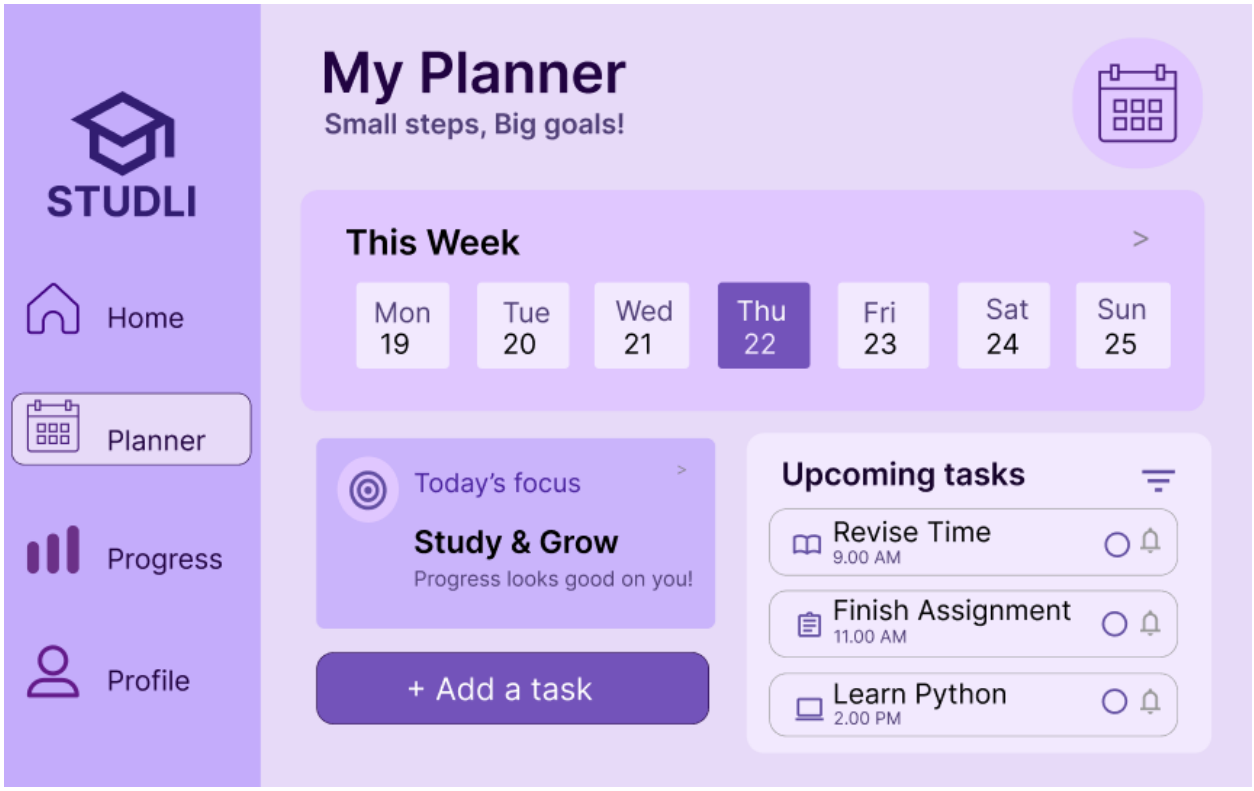


Figure 15: *Desktop Planner Frame*

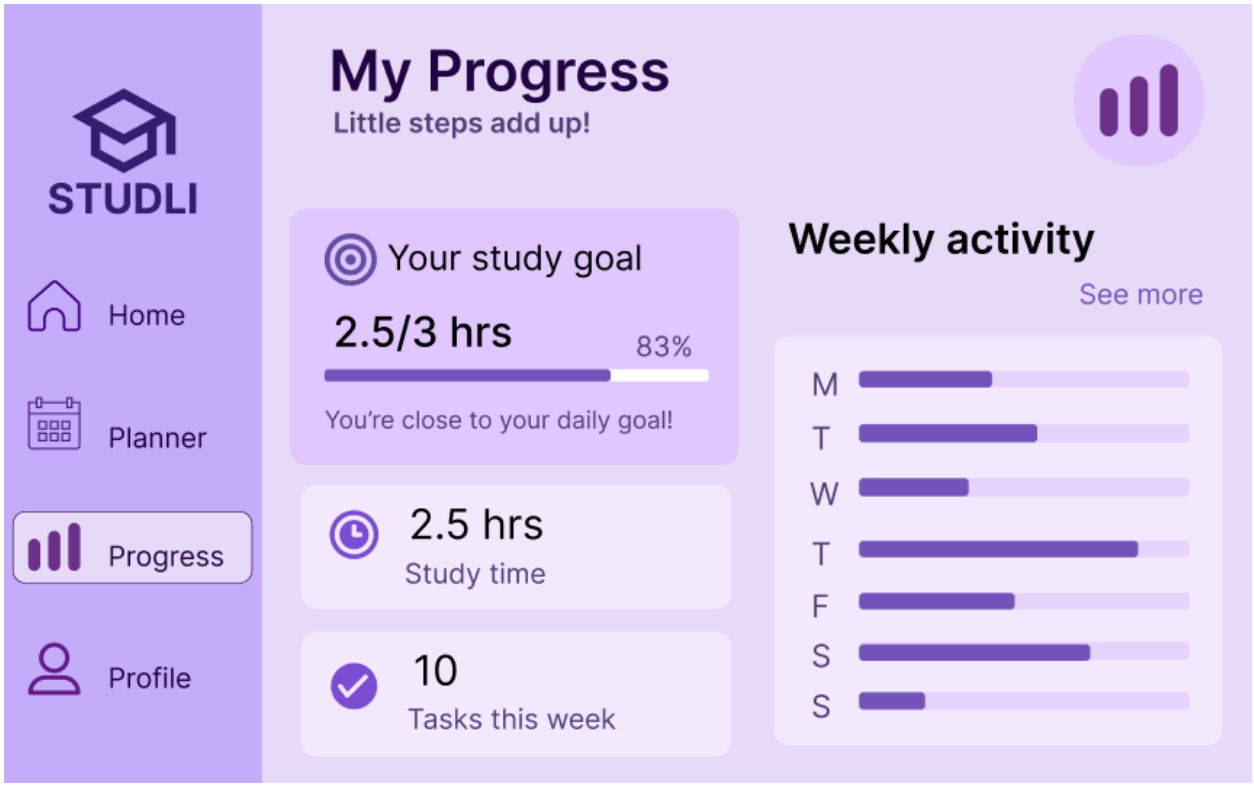


Figure 16: *Desktop Progress Frame*

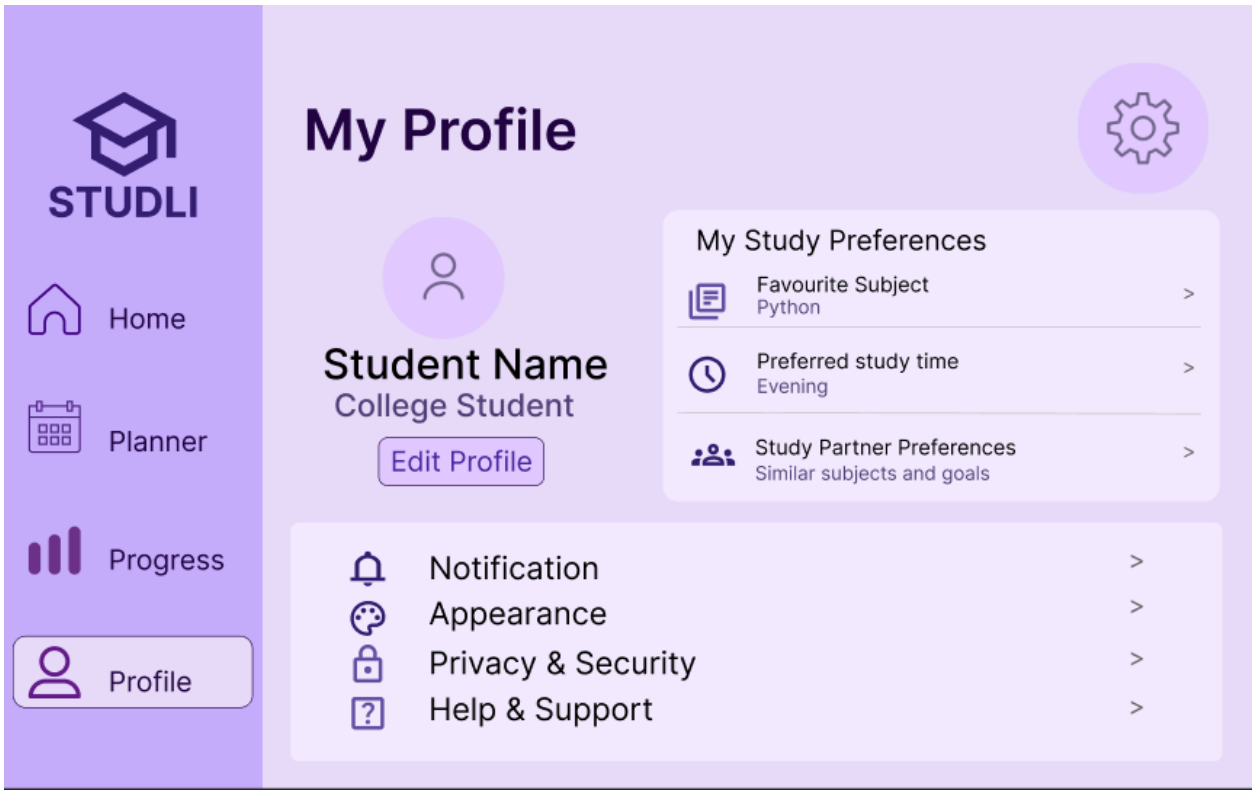


Figure 17: *Desktop Profile Frame*

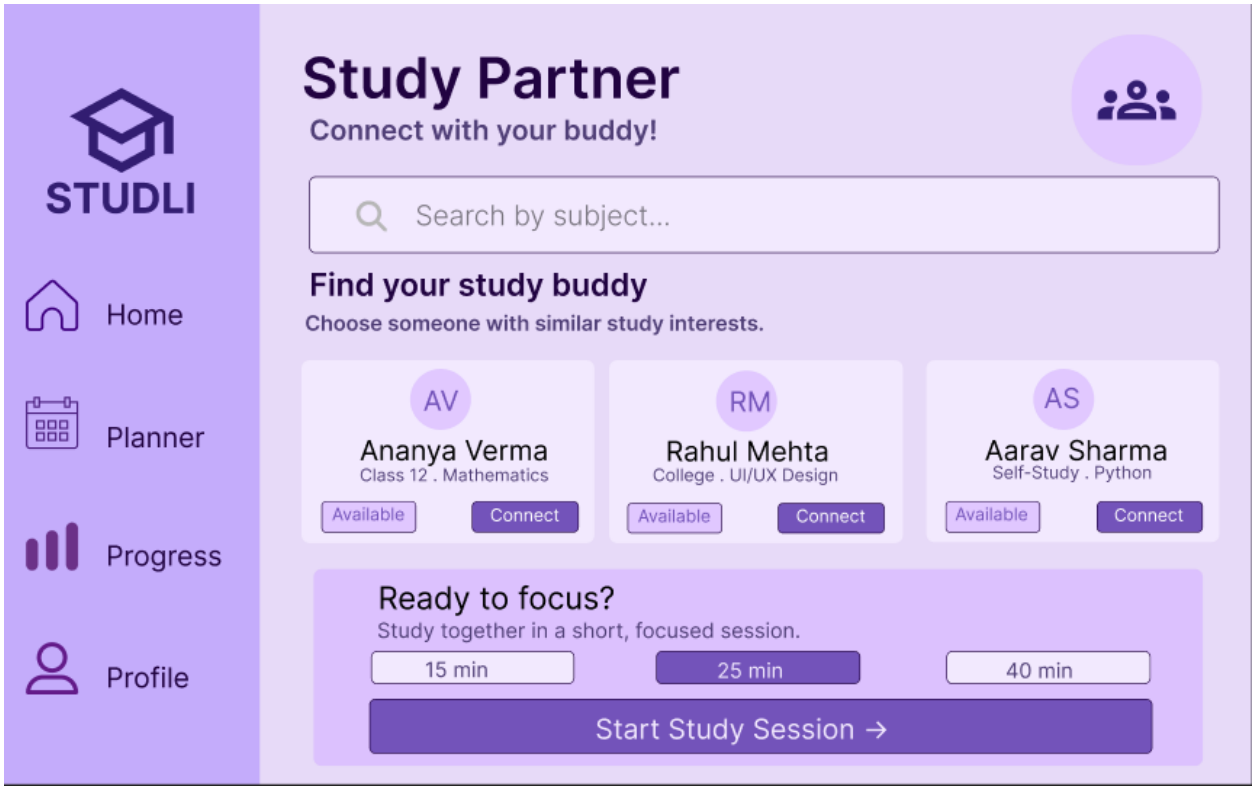


Figure 18: *Desktop Study Partner Frame*

9. Accessibility Implementation

Accessibility was considered throughout the responsive redesign of the Studli application. The objective was to improve readability, interaction and navigation while maintaining the original visual identity across mobile, tablet and desktop layouts.

9.1 Colour Contrast and Readability

Text colours were reviewed against their background colours to improve readability. Selected grey and light-purple text colours were replaced with a darker shade, "#514078", on light backgrounds. The original purple and lavender colour palette was retained to maintain visual consistency.

9.2 Touch Target Size

Interactive elements were checked to ensure that their clickable areas measured at least 44 × 44 px. This included navigation items, buttons and other interactive controls. Adequate clickable areas help make the interface easier to use, particularly on touch-screen devices.

9.3 Keyboard Focus Indicators

Visible focus indicators were added to demonstrate how users navigating through the interface with a keyboard can identify the currently selected element.

Examples from the Studli redesign:

1. Add Task Button – Focus State: A duplicate of the Add Task button was created with a 2 px dark navy outline ("#210341"). This outline makes the focused button visually distinguishable from its normal state.
2. Input Field – Focus State: A duplicate of the input field was created using the same 2 px dark navy outline. This demonstrates how a user can identify the active form field while navigating.

These examples demonstrate how visible focus indicators can improve keyboard navigation and interaction.

9.4 Typography and Responsive Font Scaling

Typography was adjusted across mobile, tablet and desktop layouts to maintain readability and visual hierarchy on different screen sizes.

- **Mobile:** Font sizes were adjusted to keep text readable within the narrower screen width and prevent unnecessary crowding.
- **Tablet:** Typography was adapted to suit the wider layout while maintaining clear headings and readable supporting text.
- **Desktop:** Font sizes and text hierarchy were adjusted to make effective use of the larger display area.

The adjustments maintained consistency between headings, subheadings, labels and supporting text. Appropriate text spacing and alignment were also considered to improve readability and content presentation.

9.5 Form Accessibility

Input fields and form controls were reviewed for clear presentation and usability. Focus-state styling was added to demonstrate keyboard interaction and make the active input field easier to identify.

9.6 Accessibility Guidelines

The improvements were guided by accessibility principles, including WCAG recommendations for readable text, sufficient contrast and keyboard accessibility. These are design-level improvements and do not represent a formal WCAG compliance audit or certification.
Overall, the accessibility updates aim to make the Studli interface clearer and easier to interact with across different devices.

10. Colour and Visual Consistency

The original Studli colour palette was retained throughout the responsive redesign to preserve the application's visual identity. The same colours were applied across mobile, tablet and desktop layouts to maintain consistency in buttons, headings, cards, backgrounds and interactive elements.

10.1 Colour Palette

The following colours were used in the redesigned interface:

Colour name	Colour	Hex Code	Usage
Dark Navy		"#210341"	Headings and focus outlines
Main Lavender		"#E7DAF8"	Main backgrounds
Card Lavender		"#E0C7FF"	Card backgrounds
Primary Purple		"#7353BA"	Buttons and progress indicators
Adjusted Text Purple		"#514078"	Supporting text and labels
White		"#FFFFFF"	Text and light interface areas

10.2 Consistency Across Devices

The same colour combinations were maintained across all three device categories. Although the layout and positioning of elements changed according to screen dimensions, the visual appearance remained consistent.

10.3 Accessibility-Related Colour Adjustments

Selected grey and light-purple text colours were changed to "#514078" on light backgrounds to improve text readability. Dark navy was also used for visible focus outlines.

These adjustments improved the distinction between text, backgrounds and interactive elements without changing the overall purple and lavender theme.

Overall, maintaining the original colour palette helped ensure that the responsive versions remained visually connected to the original Studli design.

11. Responsive Testing

Responsive testing was carried out in Figma to review how the redesigned Studli interface appeared across mobile, tablet and desktop frame sizes.
The purpose of testing was to check whether the layouts remained organised, readable and visually consistent when adapted to different screen dimensions.

11.1 Testing Specifications

Device	Width	Height	Orientation
Mobile	430 px	932 px	Portrait
Tablet	1024 px	768 px	Landscape
Desktop	1440 px	900 px	Landscape

11.2 Testing Areas

The following aspects were reviewed during the testing process:

1. Layout Alignment: Checked the positioning and alignment of interface elements across different screen sizes.
2. Navigation: Reviewed the mobile bottom navigation and the vertical sidebar used in tablet and desktop layouts.
3. Content Organisation: Checked card arrangements, sections and spacing to ensure that information remained organised.
4. Text Readability: Reviewed headings, labels and supporting text for clear presentation.
5. Interactive Elements: Checked the presentation and sizing of buttons, navigation items and input fields.
6. Screen Adaptability: Reviewed the layouts at different widths to identify crowded content, spacing issues and alignment problems.

11.3 Testing Outcome

The responsive layouts were reviewed across all three device categories. The mobile version retained its single-column structure, while the tablet and desktop versions used the available horizontal space through wider content arrangements and sidebar navigation.

The review helped confirm that the redesigned screens maintained consistent styling, organised content and clear navigation across the selected frame sizes.

12. Design Decisions and Analysis

During the responsive redesign of Studli, several design decisions were made to adapt the original mobile interface to larger screen sizes while maintaining consistency and usability.

12.1 Key Design Decisions

Design Decision	Reason	Result
Single-column mobile layout	Limited horizontal screen space	Content remains organised and easy to navigate.
Bottom navigation on mobile	Suitable for a compact portrait layout	Navigation remains accessible without taking up excessive screen space.
Vertical sidebar on tablet and desktop	Wider screens provide space for a separate navigation area	Main content receives more horizontal space.
Two-column arrangements on tablet	Better use of the available landscape layout	Related content can be displayed side by side where suitable.
Wider content areas on desktop	Larger frame dimensions allow more spacious layouts	Cards and sections have room to breathe.
Adjusted text colour	Improve readability against light backgrounds	Supporting text uses the darker shade "#514078".
Visible focus indicators	Make focused interactive elements easier to identify	Button and input field focus-state examples were created using a 2 px outline.
Consistent colour palette	Preserve the original Studli visual identity	The purple and lavender theme remains consistent across devices.

12.2 Overall Design Analysis

The redesign involved more than increasing the dimensions of the original mobile screens. The arrangement of navigation, cards and content sections was adapted according to the available screen space.

On mobile, a compact single-column structure and bottom navigation were retained. On tablet, the landscape layout allowed content to be arranged in wider sections and two columns where appropriate. On desktop, the larger frame provided additional space for wider content areas and a vertical sidebar.

Accessibility-related changes were also included through adjusted text colours, appropriately sized interactive elements and visible focus-state examples.

These decisions helped create three device-specific layouts while preserving the original application's content, colour palette and overall design identity.

13. Conclusion

The Week 4 project focused on transforming the original Studli student study planner interface into a responsive and more accessible design for mobile, tablet and desktop screens.

The redesign included a total of 17 screen designs across three device categories:

- Mobile: 7 screens at 430 × 932 px.
- Tablet: 5 screens at 1024 × 768 px
- Desktop: 5 screens at 1440 × 900 px.

The project involved adapting layouts, reorganising navigation, improving content arrangement and maintaining visual consistency across different screen sizes. Accessibility improvements included adjusted text colours, interactive elements with at least 44 × 44 px clickable areas and visible keyboard focus-state examples.

Throughout the process, the original Studli colour palette, typography, icons and content structure were retained to preserve the application's identity.

Figma was used to create and review the responsive mockups and document the design decisions.

Overall, the project provided practical experience in responsive interface design, accessibility considerations and adapting an existing application interface for multiple devices.